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Experiment - 5 : Wireshark

Aim : To capture, filter & analyse network packets using Wireshark & understand encapsulation at the protocol layers

Description :-

* Wireshark is a packet sniffer (formerly Ethereal). It passively captures packets in real time & shows them in readable form.

* It is used to troubleshoot networks, examine security problems, debug protocols & learn protocol internals.

* The window has three panes : Packet List (one line per packet), Packet Details (protocol tree) & Packet Bytes (hex dump)

* Colours :-
  TCP : Light purple
  UDP : Light blue
  Packets with errors : Black

Procedure :-

(i) Open Wireshark & select the interface (Local Area Connection).

(ii) Go to Capture → Options & set "Stop capture automatically after 100 packets".

(iii) Click "Start" capture, then Stop & save the file.

(iv) Type a filter in the filter bar & press Enter.

(v) For a flow graph, go to Statistics → Flow Graph.

(vi) Click a packet to inspect its layers.

Result: Packets were captured, filtered & analysed successfully using WireShark.

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